

Code-Layout, Readability and Reusability

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the Rising Waters: A Machine Learning Approach to Flood Prediction codebase in terms of code organization, readability, and reusability. The project follows a modular structure using Python, Flask, HTML, CSS, and Machine Learning components, making the application easy to maintain, understand, and extend.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation used throughout the code	Yes	Python PEP-8 style followed
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for templates, static files, model, and application
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names used
4	Function / Method Names	Functions are descriptively named	Yes	Functions describe their operations clearly
5	Code Comments	Inline and block comments explain log	Partial	Comments added for major sections only
6	Modular Design	Code is split into reusable functions/m	Yes	Flask routes and ML model separated
7	No Redundant Code	Duplicate or unused code is removed	Yes	Code optimized by eliminating repetition
8	Error Handling	Exceptions and errors are handled	Partial	Basic input validation and exception handling implemented

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application (app.py)	Python, Flask	Handles routing and prediction requests	High
2	Trained ML Model	XGBoost, Joblib	Used for flood prediction	High
3	HTML Templates	HTML, CSS	Home page and prediction result pages	High
4	Data Preprocessing Module	Python, Pandas	Data cleaning and feature preparation	Medium
5	Prediction Function	Python, Scikit-learn	Used for generating flood prediction results	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with a clear folder structure
Readability	5	Meaningful naming conventions and simple logic improve readability
Reusability	5	Modular design enables code reuse across the application
Documentation / Comments	4	Major sections are documented; additional comments can further improve clarity
Overall Score	4.8/5	The project demonstrates good coding practices and maintainable design suitable for future enhancements.